

BMU–GW BMS gateway

Rev B0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Isolated CAN FD + RS–485 between CTRL–C2000 and the battery BMS

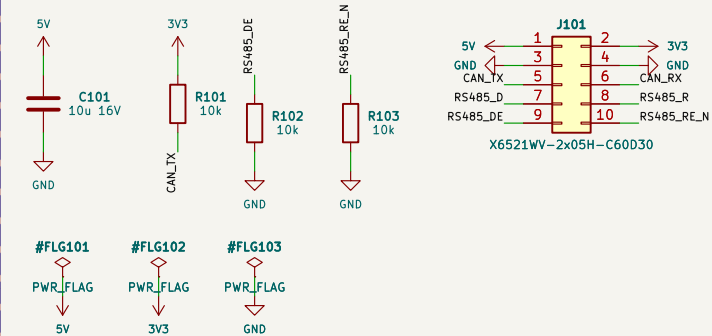
01 Isolated CAN + RS–485

Host header, push–pull isolated 5 V, ISO1042 CAN FD, ISO1410 RS–485, TVS, terminations, field connector

File: sheets/01_gateway.kicad_sch

The field side (GND_BMS) floats with the BMS communication ground; only U–isolators and the bias transformer cross the barrier (checked on every build).
The isolated 5 V is unregulated (5 V x 1.1 minus one Schottky drop): 4.8–5.5 V over a 5 V +/-5 % input, inside the 4.5–5.5 V VCC2 window of ISO1042. R preload keeps it bounded at no load.

Not bench–validated. PELV = controller side, BMS = field side. Sheet 1 Schematics authored by Chinmoy Bhuyan Barrier: 5 kVrms reinforced isolators; bias transformer 2500 VAC test (functional/basic)		
Sheet: / File: BMU–GW.kicad_sch		
Title: BMU–GW BMS gateway – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: B0
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The schematic diagram illustrates the internal components and connections of the ISO1410DWR module. It features two main integrated circuits: U103 (ISO1410DWR) and D104 (CDSOT23-SM712).

U103 (ISO1410DWR) Connections:

- VCC1 (Pin 1):** Connected to 3V3.
- D (Pin 6):** Connected to RS485_D.
- DE (Pin 5):** Connected to RS485_DE.
- RE (Pin 4):** Connected to RS485_RE_N.
- R (Pin 3):** Connected to RS485_R.
- NC (Pin 7):** Not connected (marked with an 'X').
- GND1 (Pin 2):** Connected to GND.
- GND1 (Pin 8):** Connected to GND.
- VCC2 (Pin 16):** Connected to 5V_BMS.
- A (Pin 12):** Connected to RS485_A.
- B (Pin 13):** Connected to RS485_B.
- NC (Pin 10):** Not connected (marked with an 'X').
- NC (Pin 11):** Not connected (marked with an 'X').
- NC (Pin 14):** Not connected (marked with an 'X').
- GND2 (Pin 15):** Connected to GND_BMS.
- GND2 (Pin 9):** Connected to GND_BMS.

D104 (CDSOT23-SM712) Connections:

- I01 (Pin 1):** Connected to RS485_A.
- I02 (Pin 2):** Connected to RS485_B.
- COM (Pin 3):** Connected to GND_BMS.

Other Components and Connections:

- Capacitors:** C110 (100nF 50V) and C111 (100nF 50V) are connected between 3V3 and GND, and 5V_BMS and GND_BMS, respectively.
- JP102:** A solder jumper (open by default) connects RS485_A to RS485_TA.
- R107:** A 120Ω resistor connects RS485_TA to RS485_B.

SHIELD

R108
1M

GND_BMS

SHIELD

C112
4.7nF 4.7kV

GND_BMS

J102

CAN_H	1
CAN_L	2
GND_BMS	3
RS485_A	4
RS485_B	5
SHIELD	6

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